Bioresource Technology 385 (2023) 129438

Contents lists available at ScienceDirect

Bioresource Technology

journal homepage: www.elsevier.com/locate/biortech

Natural arginine-based deep eutectic solvents for rapid microwave-assisted pretreatment on crystalline bamboo cellulose with boosting enzymatic conversion performance

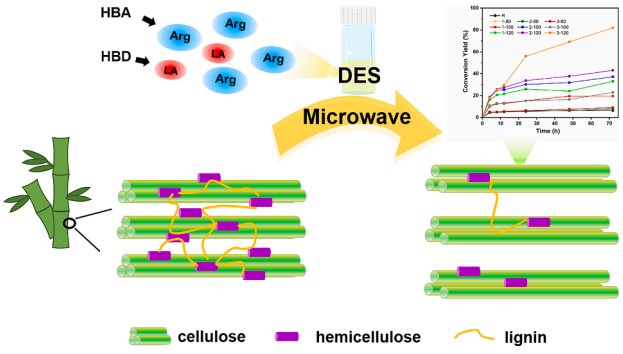
Ying Xie, Jinyi Zhao, Peng Wang, Zhe Ling *, Qiang Yong

Jiangsu Co-Innovation Center of Efficient Processing and Utilization of Forest Resources, College of Chemical Engineering, Nanjing Forestry University, Nanjing 210037, China

HIGHLIGHTS

- Novel arginine-based DES pretreatment was proposed with fast microwave assistance.
- Optimal delignification (84.8%) and saccharification (81.9%) were gained for LAA-3.
- Supramolecular structures of crystalline cellulose greatly impact on bioconversion.
- Lower viscosity and high α value of DES contribute to higher conversion yields.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Arginine-based DES
Pretreatment
Bamboo lignocellulose
Enzymatic hydrolysis
Crystalline cellulose

ABSTRACT

Development of amino acid-based natural deep eutectic solvents (DESs) pretreatment technology on lignocellulosic biomass is a promising approach to biorefinery. In this study, for arginine-based DESs with different molar ratios, the viscosity and Kamlet-Taft solvation parameters were quantified to evaluate the pretreatment performance on bamboo biomass. Further, microwave assisted DES pretreatment was eminent, evidenced by 84.8% lignin removal and increased saccharification yield (from 6.3% to 81.9%) in moso bamboo at 120 °C and ratio of 1:7 (arginine: lactic acid). Results showed degradation of lignin molecules together with release of phenolic hydroxyl units after DESs pretreatment, which is conducive to subsequent utilization. Meanwhile, DES-pretreated cellulose exhibited unique structural characteristics including destroyed crystalline region of cellulose (Crystallinity Index from 67.2% to 53.0%), decreased crystallite size (from 3.41 nm to 3.14 nm) and roughened fiber surface. Thus, arginine-based DES pretreatment has excellent potential in bamboo lignocellulose pretreatment.

* Corresponding author.
E-mail address: jjling19@njfu.edu.cn (Z. Ling).

https://doi.org/10.1016/j.biortech.2023.129438
Received 30 May 2023; Received in revised form 29 June 2023; Accepted 30 June 2023
Available online 1 July 2023
0960-8524/© 2023 Elsevier Ltd. All rights reserved.

1. Introduction

Renewable biomass has been paid considerable attention in recent years due to global warming and the depletion of fossil energy. Lignocellulose, as the most abundant renewable biomass resource, is an ideal substitute for fossil energy. It can be converted into value-added chemicals, bio-based materials and biofuels (Wang et al., 2020). However, the complex hierarchical structure and recalcitrant cross-linking chemical structure of lignocellulose are barrier for efficient hydrolysis of cellulose (Zhao et al., 2017). In order to achieve more efficient deconstruction and utilization of major components of lignocellulose, pretreatment is a significant process for biotransformation to fermentable sugars (Khan et al., 2021). Therefore, the development of pretreatment technology is the front and trend of lignocellulose biotransformation.

Recently, much effort has been devoted to develop green pretreatment technologies. Among them, deep eutectic solvents (DESs) is particularly prominent for low cost, simple synthesis and recyclability. In general, DESs use choline chloride as hydrogen bond acceptor (HBA) and acids, bases, carbohydrates and phenols as hydrogen bond donor (HBD). Due to the fact that some components are cytotoxicity, DESs are not completely innocuous and harmless solvents (Zhong et al., 2022). However, natural DESs composed of natural sources have been proved as non-toxic, biocompatible and have potential for pretreatment of biomass (Aroso et al., 2017). As the most abundant natural compound, amino acids are good resources for the synthesis of natural amino acid-based DESs, which can form the intrinsic net of hydrogen bonds with other components in specific molar ratio. Amino acid DESs (AADESs) are particularly effective in lignin depolymerization by inducing a mild acid-base catalytic environment (Alvarez-Vasco et al., 2016). Meanwhile, some amino acids acting as HBA have Bronsted acidity to break the acid-labile glycosidic linkages in hemicellulose (Chen et al., 2022). Jablonsky et al. (2019) reported that alanine/lactic acid DES (1:9) pretreatment at 60 °C for 24 h induced 23.7% of lignin removal from wheat straw. During the pretreatment of corncob lignocellulose with DES prepared from betaine and amino acids (lysine, arginine, histidine), about 50% of xylan and 60% of lignin could be removed at 60 °C for 5 h and the highest enzymatic conversion yield of 60.8% glucose was reached (Liang et al., 2020). These literatures all indicate that AADESs are green alternative solvents for lignocellulose biomass pretreatment. However, AADES pretreatments mainly focus on feedstocks with loosened cell wall structure, such as corn cob and wheat straw, while the attempt of AADESs on highly aggregated woody plants or bamboo materials was absent (Wang and Lee, 2021). Meanwhile, assessments to physicochemical changes of substrates that contributed to improved hydrolyzability and appraisals of solvatochromic parameters of AADESs are very limited. In addition, the lack of repeatability, reproducibility, long term stability furtherly hindered its development. Therefore, the limitations of the method need to be overcome and the assessment of AADESs need to be gradually expanded to other types of lignocellulosic biomass. It would also be of much necessity to explore detailed characterization of the solvatochromic parameters of AADESs and physicochemical changes of substrates during AADESs pretreatments.

DESs have been proved as one of the most promising lignocellulose pretreatment solvents, but it often takes long time for biomass treatment and fractionation (Selkälä et al., 2016). To overcome this shortage, microwave (MW) heating on biomass and the destruction of hydrogen bonds of carbohydrate by the oscillating movement of the dipoles have been reported to improve the energy efficiency in the pretreatment process (Haldar and Purkait, 2021). Therefore, the application of MW-assisted DES pretreatment shows significant potential in the field of lignocellulosic biomass biorefinery with shortened treatment time and reduced energy consumption. Liu et al. (2017) found that it took only 3 min by MW-DES (choline chloride/oxalic acid dihydrate) treatment, which received similar lignocellulose fractionation effects from oil-bath heating at 110 °C for 9 h. In addition, high delignification rate (90.83%) and the saccharification yield of glucan (96.15%) of poplar wood were achieved after MW-assisted DES (choline chloride/glycerol/$AlCl_3$) treatment at 110 °C for 10 min (Xu et al., 2022). It was reported that MW has a strong synergistic effect with DESs, which maximizes the ionic properties of DES and increases its molecular polarity to improve cellulose digestibility (Mankar et al., 2021). However, up till now, there have been very few reports pertaining to MW-assisted AADES pretreatment of lignocellulosic biomass as well as the solvent properties of AADESs during such treatments.

In this study, lactic acid /arginine (LAA) DESs were developed to pretreat moso bamboo by microwave radiation assistance for the first time, in order to achieve high-yield fermentable sugars with relatively short time. The polarity and viscosity of DESs in different molar ratios were evaluated and hinged on its pretreatment performance. Besides, the effects of DES pretreatment on cell wall morphology, lignin removal, and cellulose crystal structure were systematically investigated. The impacts of physicochemical changes of substrates on enzymatic hydrolysis of lignocellulose were elucidated. In a word, the present study provided a kind of regulative strategy for biotransformation of fermentable sugar from bamboo biomass and proposed the inner mechanisms of the novel microwave-assisted LAA pretreatment for improving the biorefinery efficiency.

2. Materials and methods

2.1. Materials and chemicals

LAA pretreatment used moso bamboo purchased from Yunnan Forestry Center. After Soxhlet extraction for 6 h (used acetone-ethanol mixture at a volume ratio of 2:1), pigments and alcohols were removed from bamboo powders (40–80 mesh). The bamboo strips were cut into transverse and radial sections in size of 10 μm by slicer (HM 325, Thermo Scientific, USA) and observed under microscope. L-arginine (Macklin Chemical Co., Shanghai, China), lactic acid, ethanol and acetone (Shanghai Chemical Reagent Co., Shanghai, China) were all analytical purity.

2.2. DES preparation and characterization

DESs were prepared by mixing lactic acid and arginine at a molar ratio of 5:1/6:1/7:1 (LAA-1, LAA-2 and LAA-3). The transparent DESs were prepared in oil bath at 80 °C. The prepared DESs were left overnight and observed for clarity. The viscosities of DESs were quantified with Brookfield kinematic viscometer (USA) at 80 °C, 100°C and 120 °C. The α, β, and π* of prepared DESs were measured (4-nitroanisole, 4-nitroaniline and Nile red as dye indicators) to indicate solvent properties (C. Florindo et al., 2017). The absorption spectrums of the mixtures (Dyes and DESs) were quantitated by UV–Vis spectrometer. The solvatochromic parameters were calculated as follow (Liang et al., 2021).

$$\pi^* = 14.57 - {}^{4270}/_{\lambda_{\max,OMe}} \quad (1)$$

$$\beta = 11.134 - {}^{3580}/_{\lambda_{\max,NH_2}} \quad (2)$$

$$\alpha = (19.9657 - 1.0241 \times \pi^* - \nu_{NR})/1.6078 \quad (3)$$

$$\nu_{NR} = {}^{1}/\left(\lambda_{\max,NR} \times 10^{-4}\right) \quad (4)$$

2.3. DES pretreatment

Moso bamboo powder and DES (solid–liquid ratio of 1:20) were mixed in Teflon microwave reactor vessel. The mixtures were heated for 10 min and held for 20 min at 600 W in MDS-8G Microwave digestion apparatus (SINEO, Shanghai). The microwave radiation temperature was set at 80 °C, 100 °C and 120 °C, respectively. After the reaction, the

solid residues were washed several times with 50% acetone solvent. The solids were repeatedly washed with water until neutral and then stored at 4 °C. Meanwhiles, the acetone in filtrate was removed by vacuum filtration and ten times volume of water was used to precipitate lignin. The lignin samples were centrifuged at 8000 rpm for 10 min and dried in vacuum dryer at 45 °C overnight.

Samples pretreated by LAA-1, LAA-2 and LAA-3 at 80 °C, 100 °C, and 120 °C were respectively marked as 1–80, 1–100 and 1–120, 2–80, 2–100 and 2–120, 3–80, 3–100 and 3–120. Raw moso bamboo sample was labeled as R.

2.4. Enzymatic hydrolysis

Cellulase (Novozymes, Celic CTec2) was used to hydrolyze solid residues under 5% (w/v) biomass loading. This work chosed 0.1 M Sodium Citrate buffer to keep the hydrolysis process at 4.8 pH and 20 FPU/g glucan as cellulase dosages. The whole process took place at 50 °C. The supernatant was taken at regular intervals and then deactivated and centrifuged. The composition of the supernatant was analyzed with high-performance liquid chromatography. The conversion yield was calculated as follows:

$$Conversion\ yield\ (\%) = \frac{Released\ glucose\ amount}{Initial\ amount\ of\ glucan\ in\ solids} \times 100\% \quad (5)$$

2.5. Characterization

2.5.1. Chemical compositional analysis

The chemical compositions of the samples were analyzed by the National Renewable Energy Laboratory (NREL). The glucose and xylose obtained after sulfuric acid hydrolysis were quantified by liquid chromatography.

The characteristic functional groups of samples were explored by Fourier transform infrared spectroscopy (FTIR) (Nicolet iS5, Thermo Scientific, USA). The wavelength range of the absorption spectrum was 500 – 4000 cm $^{-1}$. The scans were repeated three times.

2.5.2. Crystal structural analysis

The crystalline structures of the sample were analyzed by X-ray diffraction (XRD) (Ultima IV, Rigaku, Japan). The samples were scanned in the range of 5° to 50° at a rate of 5°/min (λ = 1.5419 Å). The patterns were fitted by Maud Software and calculated by following formulas:

$$CrI = \frac{A_{\text{crystal}}}{A_{\text{crystal}} + A_{amorph}} \times 100\% \quad (6)$$

$$D = \frac{K\lambda}{\beta cos[03B8]} \quad (7)$$

$$\lambda = 2dsin[03B8] \quad (8)$$

2.5.3. Microscopic surface analysis

The microscopic surface of the samples was analyzed by scanning electron microscope (SEM) (Regulus 8100, HITACHI, Japan) and confocal laser scanning microscopy (CLSM) (LSM710, ZEISS, Germany). CLSM was used to observe the lignin content in the cell wall of the untreated and pre-treated bamboo slices. Samples were directly examined for autofluorescence with laser (405 nm) as the excitation wavelength.

2.5.4. Lignin structural analysis

In HSQC NMR analysis, 50 mg lignin samples extracted from pretreatment solution were added to 1 mL dimethyl sulfoxide (DMSO)-d6. 2048 data points of F2 (1H) dimension were collected and the acquisition time was 256.1 ms. 256 incremental scans were performed 32 times with the acquisition time of 6.1 ms and a delay of 1.5 s.

2.5.5. Analytical methods

The Pearson correlation analysis was used to measure the linear correlation between variables. Heat map of the correlations was conducted using GraphPad Prism 9.0 software.

3. Results and discussion

3.1. DES characterization

The viscosity and Kamlet–Taft solvatochromic parameters of DES were investigated in the present study. Viscosity was an important parameter of DES as solvent. The viscosities of LAA were higher than that of other DESs, which was mainly attributed to the relatively large HBA size and free volume in DES system (Loow et al., 2017). However, the viscosity also depended on the mole ratio and temperature of HBA and HBD (Chang et al., 2021). As seen in Table 1, the viscosities of LAA were obviously decreased from 1018.2 (LAA-1), 713.8 (LAA-2) and 626.7 m Pa·s (LAA-3) to 129.4, 70.6 and 112.0 m Pa·s with the increasing temperature. This was because the significant reduction of hydrogen-bonding interactions in the solvents at higher temperatures (Amesho et al., 2022). Therefore, the decrement in viscosity provided more effective contacts between DES and the lignocellulosic biomass and aids lignin disintegration.

Kamlet-Taft polarity scale was an important characterization parameter for determining the polarity of DES that relate to its delignification and pretreatment capability. It quantified the hydrogen bond acidity (α), hydrogen bond basicity (β) and polarizability (π*) of DES solvents. As shown in Table 1, LAA-3 had higher π* and α values than LAA-2 and LAA-1, suggesting that LAA-3 was more effective to stabilize a charge and donate hydrogen bonds (Mankar et al., 2022). The pretreatment ability of DESs could be priorly judged by quantifying the ability of H-bond formation. It could be inferred that LAA-3 with higher lactic acid content may provide more remarkable pretreatment performance and higher lignin removal.

3.2. Chemical compositional analysis

As summarized in Fig. 1, the microwave-assisted LAA pretreatment was highly effective for fractionating moso bamboo lignocellulose. A large amount of lignin and xylan were removed by DESs and cellulose contents were increased noticeably. The HBD and HBA molar ratios of DES and the pretreatment temperature were comprehensively studied to determine the influencing factors of LAA pretreatment (Fig. 1a-c). 52.2%-84.8% of lignin and 46.4%-76.9% of xylan were removed for LAA-3 pretreatment. After LAA-1 and LAA-2 pretreatments, the removal yields all showed notable reductions. Suggesting that LAA-3 present considerable delignification capability and provide manifest hemicelluloses removal. Meanwhile, LAA-3 also showed the highest Kamlet-Taft parameter α value, the highest hydrogen-bonding donor capacity and acidity. Therefore, it could be inferred that high α value is a positive factor to improve the productivity of lignocellulosic biomass pretreatment. Previous study depicted that α value of DESs showed strong positive linear relationship with effectiveness of DES pretreatment

Table 1
The Kamlet–Taft solvatochromic parameters and viscosity of DESs.

Samples	Temperature (°C)	Viscosity (m Pa·s)	π*	α	β
LAA-1	80	1018.2	0.62	0.61	1.21
	100	360.8			
	120	129.4			
LAA-2	80	713.8	0.71	0.76	0.97
	100	147.4			
	120	70.6			
LAA-3	80	626.7	0.88	0.81	0.77
	100	223.5			
	120	112.0			

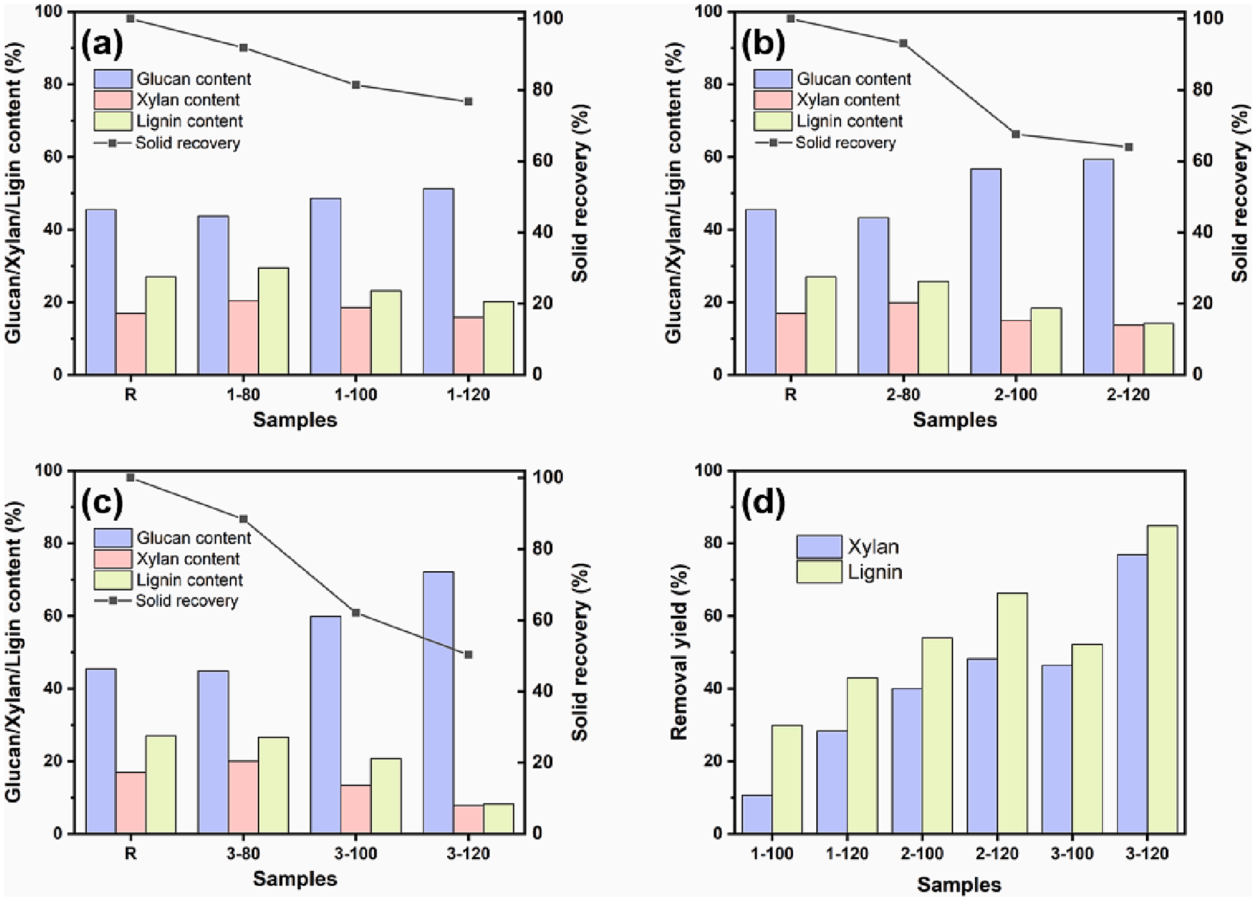


Fig. 1. The chemical composition of samples pretreated by LAA-1 (a), LAA-2 (b) and LAA-3 (c) and removal of xylan and delignification of pretreated samples (d).

(Liang et al., 2021). It had been reported that proton-catalyzed cleavage of ether, ester and glycoside linkages between lignin-carbohydrate complex during acid DES pretreatment could lead to the depolymerization and removal of lignin and xylan (Satlewal et al., 2018). Proton catalysis depended on the dissociation of lactic acid, which was related to the acidity of DES. Therefore, the higher acidity and content of lactic acid in the present DES system were more favorable for the selective removal of lignin and xylan.

When the temperature elevated from 80 °C to 120 °C, the increase of cellulose content and the decrease of lignin and xylan content were also observed (Fig. 1d). The results showed that higher LAA pretreatment temperature could markedly remove lignin and xylan, which was consistent with the previous study (Ma et al., 2021). Li et al. (2019) suggested that glycine/lactic acid DES could remove 58% lignin from bamboo at 120 °C for 12 h by traditional heating method. Under the same temperature, the MW-assisted LAA pretreatment only needed 30 min to obtain the effective pretreatment achievement. The removal rate of lignin was similar to that of the traditional choline chloride/lactic acid DES system (Chen and Wan, 2018). This rapid and efficient treatment technology would help to save energy and production costs in the bio-refining industry and be conducive to downstream enzymatic hydrolysis.

FT-IR analysis was carried out to investigate the changes in chemical structure of moso bamboo samples (see supplementary materials). The peaks at 897, 1030, 1423 and 2896 cm^{-1} were attributed to the C-H, C-O, O-H and C-H deformation vibration of cellulose. The skeletal vibrations of the aromatic rings occurred at 1456, 1511 and 1603 cm^{-1}. Compared with untreated moso bamboo, the decrease of these three absorption bands of samples pretreated by LAA suggested that lignin depolymerized during pretreatment. The broad absorption peak around 3334 cm^{-1} was assigned to the stretch of O-H group in hemicellulose, cellulose and lignin. The absorption peak around 1367 cm^{-1} were assigned to the partial acetylation of O-H groups in hemicellulose (Ji et al., 2017). The peak at 1729 and 1240 cm^{-1} were derived from carbonyl stretching, including partial acetylation of ether bonds and hydroxyl groups in lignin and hemicellulose (Li et al., 2021). The absorption bands at 1240 cm^{-1} almost disappeared for samples 1–120, 2–120 and 3–120, which indicated the removal of lignin and xylan during pretreatment. These results were similar to those in the compositional analysis.

3.3. Morphology analysis

The morphological changes of bamboo lignocellulose samples were observed by SEM techniques (see supplementary materials). SEM image of untreated samples showed large size and rough surface. There was no obvious difference between the sample treated by LAA-1 and the untreated sample in assessing surface roughness. LAA-1 treated samples showed limited variations as compared to R, whereas samples treated by LAA-2 and LAA-3 exhibited the characteristics of minor fiber bundles, smooth surface and loose fiber. This is due to the drastic removal of lignin and xylan and retention of cellulose structure by LAA. When the pretreatment temperature reached 120 °C, the compact and ordered fiber bundle of the samples have been split into smaller fasciculus and irregular aggregates. This revealed that cellulose structure suffered considerable rupture at high temperature. What is more, the destruction of morphological structure of cellulose and physical barrier of moso bamboo lignocellulose could expose more surface area and adsorption sites of the samples and improve fermentable sugar yields (Li and Zheng, 2018).

CLSM could investigate lignin distribution in treated samples on the microscopic level (see supplementary materials). The fluorophores in lignin structure (such as phenylcoumarans and stilbenes) can generate autofluorescence at a certain excitation wavelength. Therefore, CLSM could semi-qualitatively analyze the changes of lignin in cell wall of moso bamboo after LAA pretreatment and visualize the morphology. The image revealed that fibers of the raw bamboo samples were well arranged with the highest auto-fluorescence. LAA pretreatment destroyed the morphology of the samples, which was consistent with the SEM results. Figure indicated that the autofluorescence intensity of the samples after pretreatment obviously decreased. The fluorescence decay behavior relied strongly on lignin removal of the samples during pretreatment, indicating that LAA partially removed or degraded lignin, which corresponded to the results of analysis of chemical composition. At the same time, the autofluorescence intensity of samples after LAA-3 pretreatment was weakest. It showed that LAA-3 led to a greater dissociation of some linkages between lignin and cellulose (Xue et al., 2020). This was consistent with the conclusion of DES characterization.

3.4. Crystal structural analysis

XRD was used for evaluating the crystal structure of pretreated bamboo samples (see supplementary materials). Four diffraction peaks at around 2θ = 14.9°, 16.7°, 22.5°and 34.5° of XRD patterns were assigned to (1–10), (110), (200) and (004) lattice planes of crystalline cellulose I_β. This suggested that the bamboo lignocellulose retains typical cellulose I_β allomorphs after LAA pretreatment. D-Spacings and crystallite sizes perpendicular to hydrophobic (200) plane deserve to be discussed, due to the readily adsorption of cellulose enzymes onto this hydrophobic surfaces (Selkälä et al., 2016). D-Spacing was basically unchanged and the crystallite size decreases from 3.43 nm to 3.14 nm (see supplementary materials). Fig. 2 showed that the decrease in the crystallite size of the (200) plane contributed to an increased number of cellulose chains exposed and active sites (Sang et al., 2022). The crystallinity index (CrI) of 1–100 and 2–120 increased from 67.2% to 67.3% −68.2%. The results conformed to the recently reported study (Duan et al., 2023). The slightly increased CrI resulted from the removal of amorphous components (lignin and xylan). However, with the increase of acid concentration, the crystalline region of cellulose swelled gradually, which facilitated the reaction between DES and cellulose chains in the crystalline region. The CrI of the remaining samples showed a decreasing trend. It depicted that LAA noticeably decrystallized cellulose and provided high accessibility of cellulose, which also confirmed by previous findings (Morán-Aguilar et al., 2022). Indeed, DESs could diffuse within the cellulose backbone and wreck the hydrogen-bond network with alteration of cellulose polymorph, which favors amorphization of cellulose. Moreover, the deconstruction of hydrogen-bond networking and decrystallization of cellulose to form amorphous structure by LAA would be conducive to the enzymatic hydrolysis of cellulose.

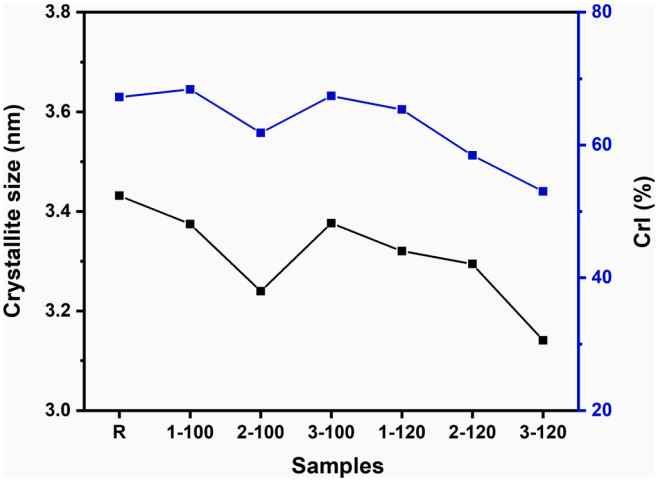


Fig. 2. Crystallite sizes perpendicular to (200) lattice plane for samples and crystallinity of the samples.

3.5. Enzymatic hydrolysis

As shown in Fig. 3, the cellulase digestibility of bamboo was 6.3% while the value increased to 19.5% −37.2% after LAA pretreatment at 100 °C. After pretreatment at 120 °C, the digestibility of cellulose significantly promoted to 81.9%, which was 13 times higher than that of raw materials. The increase of enzymatic hydrolysis rate was related to the removal of lignin and xylan, which impeded cellulase hydrolysis by acting as a physical barrier and limiting the entry of enzymes or by non-productive enzyme binding (Sun et al., 2015). DESs could effectively break down the physical barrier and improve the accessibility of cellulase, therefore increasing the glucose yield. Digestibility of pretreated samples increased with higher pretreatment temperature and lactic acid content, which reduced the viscosity of the solution. LAA had a higher viscosity than other DESs (Chourasia et al., 2022), particularly at 80 °C. Excessively viscous DESs could obstruct mass transport phenomena during the pretreatment of lignocellulose (Savi et al., 2019). This explained the low removal rate of lignin and the low yield of enzymatic hydrolysis from LAA treatment at 80 °C. With the increase of pretreatment temperature, the reaction of samples more completed due to the decrease of viscosity, especially at 120 °C.

3.6. Lignin characteristics

2D-HSQC NMR revealed an insight into lignin characterization and variation of intermolecular linkages during different DES pretreatment (see supplementary materials). Figure showed typical lignin signals and evident substructures were annotated according to the references (Xu et al., 2019).

The primary signal of β-O-4 linkages at α, β, and γ position of the lignin macromolecule (Aα, Aβ, Aγ) and the signal of acylated β-O-4 connection (A') were identified in side-chain region. Moreover, the signals of resinol structures formed by β-β/α-O-γ/γ-O-α linkages (B), phenylcoumarane structures formed by β-5/ α-O-4 linkages (C) and methoxy groups were clearly recognized. Owing to the preponderance of γ-acylation of lignin, there were some β-β units in bamboo in the form of tetrahydrofuran (THF) structures (Kellock et al., 2019). As reflected in

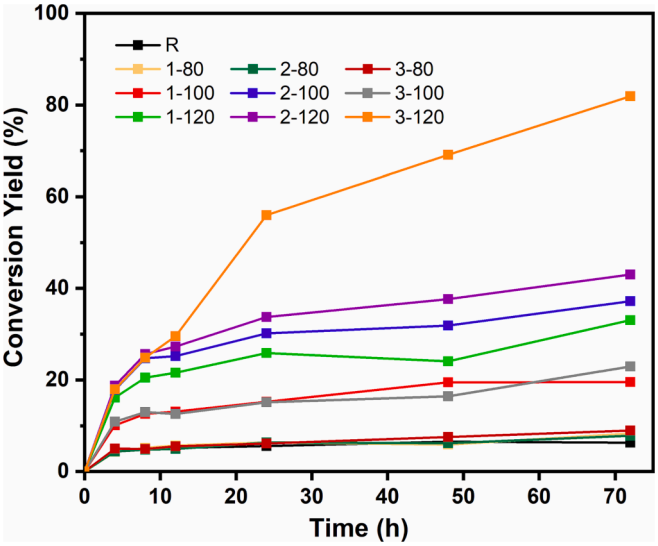


Fig. 3. Enzymatic hydrolysis of pretreated moso bamboo samples.

the aromatic regions of lignin samples, it was easy to find signals from ferulate (FA) and p-coumaric acid (PCAα, PCAβ and $PCA_{2,6}$), which indicated the dissolution of both FA and pCA during DES pretreatments. FA was an important component of lignin-carbohydrate complex (LCC), which was linked to lignin by ether bonds and hemicellulose by ester bonds. The detected FA signal meant that part of the LCC linkages were broken during DES pretreatments (Xie et al., 2020). Furthermore, the signals of PCA after LAA pretreatments suggested that DES pretreatments could decompose the end groups of p-coumaric acid (Liu et al., 2018).

In bamboo lignin, cross-signals from syringyl (S), guaiacyl (G) and p-hydroxyphenyl (H) were obviously distinguished. The contents of S and G were high, whereas those of H were low. The S/G ratios of 1–120, 2–120, 3–120 were 1.917, 1.708 and 0.966. This indicated that DESs degraded lignin seriously, including degradation of S-type and G-type lignin fragments. Meanwhile, fewer G-type implied that DES pretreatment resulted in condensation of lignin in different degrees (Xie et al., 2021). In addition, S-type units were conducive to the formation of β-O-4 bonds, so the content of lactic acid in DES was related to the breaking of β-O-4 bonds. The release of phenolic hydroxyl units caused by the break of β-O-4 bond is beneficial to the subsequent functionalization of lignin (Xu et al., 2022).

3.7. Factors impacting cellulase digestibility

To definitely illustrate the effect of substrate-related factors and DES viscosity on enzymatic hydrolysis of bamboo lignocellulose, heat map of Pearson correlation coefficient was shown in Fig. 4. Negative correlations were found between glucose yield and CrI and crystallite size of cellulose correlation coefficient (correlation coefficients were 0.954 and 0.942). Amongst, CrI was an indispensable limiting factor related to cellulase digestibility, affected by the degradation of polysaccharides and the breaking of dense hydrogen-bond network. During LAA pretreatment, the supramolecular structures of cellulose chains in the crystalline region were destroyed and the CrI of cellulose was reduced. In addition, the decrease in CrI induced low degree of polymerization (DP), high-surface area contact and porosity of cellulose, which was beneficial to the accessibility to cellulase (Xu et al., 2020). Therefore, the reduction of CrI enhanced subsequent lignocellulose enzymatic saccharification. Meanwhile, acid hydrolysis shrunk the crystallite size, providing a large surface area of the cellulose samples and enhancing the hydrolysis rate by cellulase (Wada et al., 2010). Besides, xylan removal and lignin removal showed the strong linear correlation with glucose yield with the absolute value of Pearson's r of 0.863 and 0.905. This was also consistent with the view that the removal of lignin and xylan contribute to the accessibility of the polysaccharides and accelerate the enzymatic hydrolysis of cellulose (Su et al., 2023).

Furthermore, the viscosities of DESs were negatively correlated with glucose yield, xylan and lignin removal. This was due to a higher mobility of free species within the DES with low viscosity at high temperature and the hydrogen bond network was easier to break down and recombine (Catarina Florindo et al., 2017). Therefore, the viscosity of DES was also a vital factor affecting the effect of pretreatment. In addition, the Kamlet-Taft α parameters of LAA were positively correlated with the delignification efficiency, removal xylan and enzymatic digestibility of LAA pretreated bamboo residues. High lactic acid concentration might affect the proton-donating capacity of the DES, which contributed to the release of proton to cause bond breakage in lignocellulose and catalytic depolymerization of lignin (Kang et al., 2023). Correlation study showed that low viscosity and high Kamlet-Taft α parameters have played a significant role in pretreatment performance of DES solutions.

4. Conclusions

Rapid microwave-assisted arginine-based DESs pretreatment was developed to optimize the enzymatic saccharification of moso bamboo lignocellulose. LAA could enhance lignocellulose digestibility up to 81.9%. The highest lignin and xylan removal reached 84.8% and 76.9%. Meanwhile, LAA caused disruption of the crystalline region, decrease in crystallite size and release of phenolic hydroxyl units. Furthermore, LAA-3 with low viscosity and high α value (α = 0.81) was proficient in

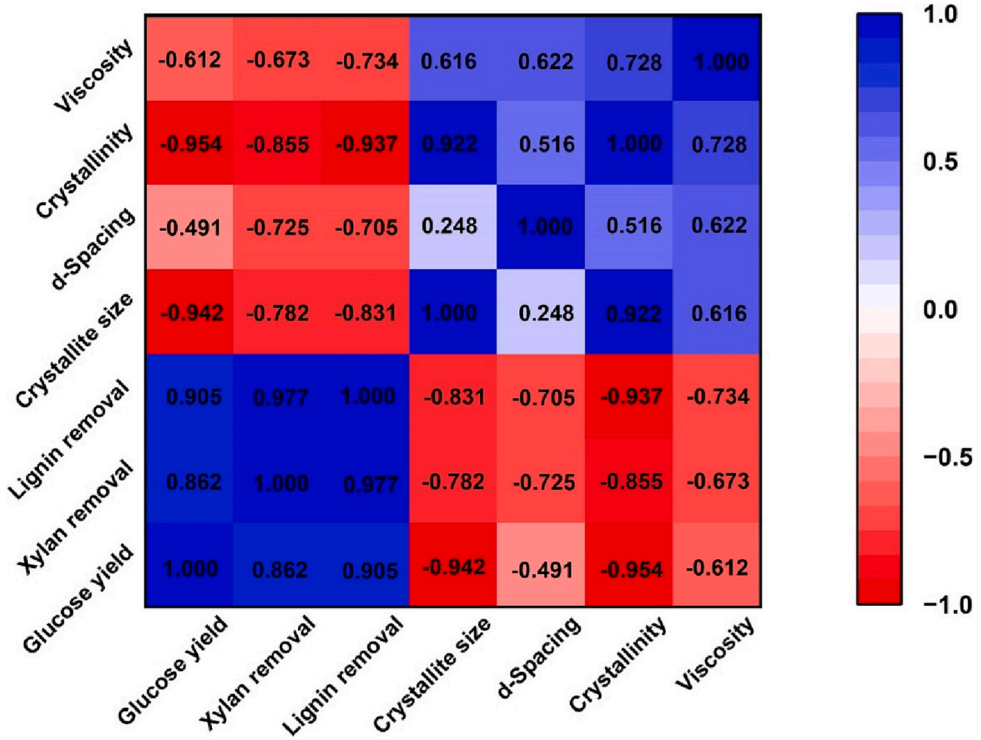


Fig. 4. Correlations between viscosity, crystallinity, d-Spacing, crystallite size, lignin removal and enzymatic hydrolysis of pretreated samples.

reaction efficiency. In summary, the proposed green LAA DESs as well as the multi structural variations of lignocellulose provided new understandings on design and selection of DES solvents for lignocellulose pretreatment.

CRediT authorship contribution statement

Ying Xie: Conceptualization, Data curation, Formal analysis, Writing – review & editing. **Jinyi Zhao:** Methodology. **Peng Wang:** Formal analysis. **Zhe Ling:** Funding acquisition, Supervision, Writing – review & editing. **Qiang Yong:** Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgements

The authors acknowledge the funding support from the National Natural Science Foundation for Youth (32001270) and Natural Science Foundation of Jiangsu Province for Youth (BK20200797).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2023.129438.

References

Alvarez-Vasco, C., Ma, R., Quintero, M., Guo, M., Geleynse, S., Ramasamy, K.K., Wolcott, M., Zhang, X., 2016. Unique low-molecular-weight lignin with high purity extracted from wood by deep eutectic solvents (DES): a source of lignin for valorization. Green Chem. 18, 5133–5141.

Amesho, K.T.T., Lin, Y.-C., Mohan, S.V., Halder, S., Ponnusamy, V.K., Jhang, S.-R., 2022. Deep eutectic solvents in the transformation of biomass into biofuels and fine chemicals: a review. Environ. Chem. Lett. 1–48.

Aroso, I.M., Paiva, A., Reis, R.L., Duarte, A.R.C., 2017. Natural deep eutectic solvents from choline chloride and betaine – physicochemical properties. J. Mol. Liq. 241, 654–661.

Chen, Z., Wan, C., 2018. Ultrafast fractionation of lignocellulosic biomass by microwave-assisted deep eutectic solvent pretreatment. Bioresour. Technol. 250, 532–537.

Chen, Z., Wang, Y., Cheng, H., Zhou, H., 2022. Hemicellulose degradation: an overlooked issue in acidic deep eutectic solvents pretreatment of lignocellulosic biomass. Ind. Crop. Prod. 187, 115335.

Chourasia, V.R., Bisht, M., Pant, K.K., Henry, R.J., 2022. Unveiling the potential of water as a co-solvent in microwave-assisted delignification of sugarcane bagasse using ternary deep eutectic solvents. Bioresour. Technol. 351, 127005.

Duan, C., Feng, X., Tian, C., Tian, G., Nie, S., 2023. Acidic deep eutectic solvent treatment for viscosity control and reactivity enhancement of bamboo dissolving pulp. Cellul. 30, 2097–2109.

Florindo, C., McIntosh, A.J.S., Welton, T., Branco, L.C., Marrucho, I.M., 2017a. A closer look into deep eutectic solvents: exploring intermolecular interactions using solvatochromic probes. Phys. Chem. Chem. Phys. 20, 206–213.

Florindo, C., Oliveira, M.M., Branco, L.C., Marrucho, I.M., 2017b. Carbohydrates-based deep eutectic solvents: thermophysical properties and rice straw dissolution. J. Mol. Liq. 247, 441–447.

Haldar, D., Purkait, M.K., 2021. A review on the environment-friendly emerging techniques for pretreatment of lignocellulosic biomass: Mechanistic insight and advancements. Chemosphere 264, 128523.

Jablonsky, M., Haz, A., Majova, V., 2019. Assessing the opportunities for applying deep eutectic solvents for fractionation of beech wood and wheat straw. Cellul. 26, 7675–7684.

Ji, H., Song, Y., Zhang, X., Tan, T., 2017. Using a combined hydrolysis factor to balance enzymatic saccharification and the structural characteristics of lignin during pretreatment of Hybrid poplar with a fully recyclable solid acid. Bioresour. Technol. 238, 575–581.

Kang, X., Deng, C., Shinde, R., Lin, R., Murphy, J.D., 2023. Renewable deep eutectic solvents pretreatment improved the efficiency of anaerobic digestion by lignin extraction from willow. Energ. Conver. Manage. 288, 117115.

Kellock, M., Maaheimo, H., Marjamaa, K., Rahikainen, J., Zhang, H., Holopainen-Mantila, U., Ralph, J., Tamminen, T., Felby, C., Kruus, K., 2019. Effect of hydrothermal pretreatment severity on lignin inhibition in enzymatic hydrolysis. Bioresour. Technol. 280, 303–312.

Khan, M.F.S., Akbar, M., Xu, Z., Wang, H., 2021. A review on the role of pretreatment technologies in the hydrolysis of lignocellulosic biomass of corn stover. Biomass Bioenergy 155, 106276.

Li, L., Yu, L., Wu, Z., China, P.R., 2019. Delignification of poplar wood with lactic acid-based deep eutectic solvents. Wood Res., 64.

Li, H., Li, X., You, T., Li, D., Nawaz, H., Zhang, X., Xu, F., 2021. Insights into alkaline choline chloride-based deep eutectic solvents pretreatment for Populus deltoides: Lignin structural features and modification mechanism. Int. J. Biol. Macromol. 193, 319–327.

Li, X., Zheng, Y., 2018. Investigation of dynamic changes of substrate features on enzymatic hydrolysis of lignocellulosic biomass. Ind. Crop. Prod. 111, 414–421.

Liang, Y., Duan, W., An, X., Qiao, Y., Tian, Y., Zhou, H., 2020. Novel betaine-amino acid based natural deep eutectic solvents for enhancing the enzymatic hydrolysis of corncob. Bioresour. Technol. 310, 123389.

Liang, X., Zhu, Y., Qi, B., Li, S., Luo, J., Wan, Y., 2021. Structure-property-performance relationships of lactic acid-based deep eutectic solvents with different hydrogen bond acceptors for corn stover pretreatment. Bioresour. Technol. 336, 125312.

Liu, Y., Chen, W., Xia, Q., Guo, B., Wang, Q., Liu, S., Liu, Y., Li, J., Yu, H., 2017. Efficient cleavage of lignin-carbohydrate complexes and ultrafast extraction of lignin oligomers from wood biomass by microwave-assisted treatment with deep eutectic solvent. ChemSusChem 10, 1692–1700.

Liu, C., Wang, X., Lin, F., Zhang, H., Xiao, R., 2018. Structural elucidation of industrial bioethanol residual lignin from corn stalk: a potential source of vinyl phenolics. Fuel Process. Technol. 169, 50–57.

Loow, Y.-L., New, E.K., Yang, G.H., Ang, L.Y., Foo, L.Y.W., Wu, T.Y., 2017. Potential use of deep eutectic solvents to facilitate lignocellulosic biomass utilization and conversion. Cellul. 24, 3591–3618.

Ma, C.-Y., Peng, X.-P., Sun, S., Wen, J.-L., Yuan, T.-Q., 2021. Short-time deep eutectic solvents pretreatment enhanced production of fermentable sugars and tailored lignin nanoparticles from abaca. Int. J. Biol. Macromol. 192, 417–425.

Mankar, A.R., Pandey, A., Modak, A., Pant, K.K., 2021. Pretreatment of lignocellulosic biomass: a review on recent advances. Bioresour. Technol. 334, 125235.

Mankar, A.R., Pandey, A., Pant, K.K., 2022. Microwave-assisted extraction of lignin from coconut coir using deep eutectic solvents and its valorization to aromatics. Bioresour. Technol. 345, 126528.

Morán-Aguilar, M.G., Calderón-Santoyo, M., de Souza Oliveira, R.P., Aguilar-Uscanga, M.G., Domínguez, J.M., 2022. Deconstructing sugarcane bagasse lignocellulose by acid-based deep eutectic solvents to enhance enzymatic digestibility. Carbohydr. Polym. 298, 120097.

Sang, S., Zhuang, X., Chen, H., Qin, Y., Cao, J., Fan, F., Lan, T., 2022. Effect of supramolecular structural changes during the crystalline transformation of cellulose on its enzymatic hydrolysis. Ind. Crop. Prod. 180, 114687.

Satlewal, A., Agrawal, R., Bhagia, S., Sangoro, J., Ragauskas, A.J., 2018. Natural deep eutectic solvents for lignocellulosic biomass pretreatment: Recent developments, challenges and novel opportunities. Biotechnol. Adv. 36, 2032–2050.

Savi, L.K., Carpiné, D., Waszczynskyj, N., Ribani, R.H., Haminiuk, C.W.I., 2019. Influence of temperature, water content and type of organic acid on the formation, stability and properties of functional natural deep eutectic solvents. Fluid Phase Equilib. 488, 40–47.

Selkälä, T., Sirviö, J.A., Lorite, G.S., Liimatainen, H., 2016. Anionically stabilized cellulose nanofibrils through succinylation pretreatment in urea-lithium chloride deep eutectic solvent. ChemSusChem 9, 3074–3083.

Su, Y., Wang, P., Lai, C., Huang, C., Ling, Z., Yong, Q., 2023. Revealing key factors influencing enzymatic digestibility of hydrothermally pretreated poplar in comparison with corn stover. Ind. Crop. Prod. 194, 116297.

Sun, F.F., Wang, L., Hong, J., Ren, J., Du, F., Hu, J., Zhang, Z., Zhou, B., 2015. The impact of glycerol organosolv pretreatment on the chemistry and enzymatic hydrolyzability of wheat straw. Bioresour. Technol. 187, 354–361.

Wada, M., Ike, M., Tokuyasu, K., 2010. Enzymatic hydrolysis of cellulose I is greatly accelerated via its conversion to the cellulose II hydrate form. Polym. Degrad. Stab. 95, 543–548.

Wang, W., Lee, D.-J., 2021. Lignocellulosic biomass pretreatment by deep eutectic solvents on lignin extraction and saccharification enhancement: a review. Bioresour. Technol. 339, 125587.

Wang, C., Zhang, X., Liu, Q., Zhang, Q., Chen, L., Ma, L., 2020. A review of conversion of lignocellulose biomass to liquid transport fuels by integrated refining strategies. Fuel Process. Technol. 208, 106485.

Xie, D., Gan, T., Su, C., Han, Y., Liu, Z., Cao, Y., 2020. Structural characterization and antioxidant activity of water-soluble lignin-carbohydrate complexes (LCCs) isolated from wheat straw. Int. J. Biol. Macromol. 161, 315–324.

Xie, J., Xu, J., Cheng, Z., Zhu, S., Wang, B., 2021. Phosphotungstic acid assisted with neutral deep eutectic solvent boost corn straw pretreatment for enzymatic saccharification and lignin extraction. Ind. Crop. Prod. 172, 114058.

Xiong Chang, X., Mujawar Mubarak, N., Ali Mazari, S., Sattar Jatoi, A., Ahmad, A., Khalid, M., Walvekar, R., Abdullah, E.C., Karri, R.R., Siddiqui, M.T.H., Nizamuddin, S., 2021. A review on the properties and applications of chitosan, cellulose and deep eutectic solvent in green chemistry. J. Ind. Eng. Chem. 104, 362–380.

Xu, L.-H., Ma, C.-Y., Zhang, C., Liu, J., Peng, X.-P., Yao, S.-Q., Min, D.-Y., Yuan, T.-Q., Wen, J.-L., 2022. Ultrafast fractionation of wild-type and CSE down-regulated poplars by microwave-assisted deep eutectic solvents (DES) for cellulose

bioconversion enhancement and lignin nanoparticles fabrication. Ind. Crop. Prod. 176, 114275.
Xu, H., Peng, J., Kong, Y., Liu, Y., Su, Z., Li, B., Song, X., Liu, S., Tian, W., 2020. Key process parameters for deep eutectic solvents pretreatment of lignocellulosic biomass materials: a review. Bioresour. Technol. 310, 123416.
Xu, G., Shi, Z., Zhao, Y., Deng, J., Dong, M., Liu, C., Murugadoss, V., Mai, X., Guo, Z., 2019. Structural characterization of lignin and its carbohydrate complexes isolated from bamboo (Dendrocalamus sinicus). Int. J. Biol. Macromol. 126, 376–384.
Xue, Y., Qiu, X., Ouyang, X., 2020. Insights into the effect of aggregation on lignin fluorescence and its application for microstructure analysis. Int. J. Biol. Macromol. 154, 981–988.
Zhao, X., Qi, F., Liu, D., 2017. Hierarchy nano- and ultrastructure of lignocellulose and its impact on the bioconversion of cellulose. Nanotechnology for Bioenergy and Biofuel Production 117–151.
Zhong, Y., Wu, J., Kang, H., Liu, R., 2022. Choline hydroxide based deep eutectic solvent for dissolving cellulose. Green Chem. 24, 2464–2475.